

Institute of Economic Studies – Charles University

Advanced Econometrics – MSc. Finance and Data Analytics

Mortgage Credit and Unequal Outcomes

Assessing Racial and Ethnic Disparities in Mortgage Loan Approvals

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Abstract

This paper investigates whether mortgage approval probabilities differ systematically across racial and ethnic groups. We use data from the Home Mortgage Disclosure Act (HMDA) for New York State in 2024. We estimate logit and probit models and condition on core underwriting, applicant, property, and area controls, used by lenders at the time of decision. We find statistically significant negative average marginal effects for most racial and ethnic minority groups. Our robustness checks confirm the improvement in model fit and explanatory power from the addition of race and ethnicity as explanatory variables. While our findings are consistent with past research, we similarly cannot conclude on the presence of discrimination in mortgage lending due to the possibility of omitted variable bias, such as the exclusion of unobserved credit scores and economic factors which could be correlated with the controls. As such, the results from this study should be interpreted as conditional on the factors considered, and not as causal.

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1. Introduction

Why, despite decades of legislation aimed at ensuring equal access to credit, such as the Fair Housing Act, do mortgage approval rates continue to differ across racial and ethnic groups? In the United States, the majority of home buyers rely on mortgage financing when purchasing a home (National Association of Realtors, 2025). Given the central role of housing credit in promoting homeownership, persistent differences in credit approval outcomes raise important socioeconomic and policy concerns. For example, Popick (2022) finds that even among approved applicants, minority groups face higher borrowing costs. At the same time, the literature highlights that differences in approval rates may partially reflect underlying disparities in economic characteristics rather than discrimination (Bostic, 1996; Munnell et al., 1996). Munnell et al. (1996), for example, emphasize that while it is difficult to infer causal discrimination from mortgage application data, minority applicants may nevertheless be discouraged during earlier stages of the lending process. As such, examining the relationship of race, ethnicity and mortgage applications throughout time, as to evaluate whether these dynamics persist or change, remains relevant. Consequently, this provides insight into whether laws and regulations aimed at promoting equal access to credit have a positive effect on the mortgage lending market. Accordingly, the main research question guiding our assessment is:

“Do mortgage approval probabilities exhibit systematic differences across race and ethnicity after conditioning on common observed underwriting, applicant, property, and area characteristics?”

To address this question, our paper uses recent HMDA data for New York State and estimates logit and probit models. Probabilities are evaluated through average marginal effects and robustness tests across model specifications. The rest of the paper is organised as follows: Section 2 reviews the existing literature and motivates the empirical question. Section 3 describes the data, variable selection, and presents summary statistics. Section 4 outlines the methodology, econometric models, estimation approach, and inference procedure. Section 5 presents our results and discusses marginal effects, as well as hypothesis testing. Section 6 concludes with a summary of findings, implications, and limitations.

2. Literature review

On average, minority mortgage applicants appear to have less wealth and higher loan-to-value ratios than white applicants, which largely accounts for the difference in acceptance rates (Munnell et al., 1996). Furthermore, Munnell et al. (1996) emphasize that key determinants of creditworthiness, such as credit scores, which could be correlated with race, are omitted from the Home Mortgage Disclosure Act (HMDA) data, therefore making conclusions about race’s role in mortgage lending limited. Similarly, Bostic (1996) argues that to infer the existence of discrimination among minority applicants, one must prove that observed differences are independent of economic factors. Nevertheless, Cherian (2014) argues that it remains important to

test whether race plays a statistically significant role in explaining the outcome of mortgage applications, conditional on a set of common observed underwriting, applicant, property and area characteristics. Given the limited scope of available data, our analysis, similar to previous research, should be interpreted as conditional on observed covariates rather than causal. Although the relationship of race, ethnicity and mortgage approval rates has been widely studied over the past three decades, it remains relevant to revisit the topic using recent data and focus on specific regions, such as New York State in 2024. Hence, we outline the following null hypothesis:

H₀: *Race and ethnicity have no effect on mortgage approval probabilities conditional on observed underwriting, applicant, property, and area characteristics.*

While prior research stresses that omitted creditworthiness variables limit causal interpretation, they also suggests that conditional tests remain informative. Accordingly, this paper revisits the role of race and ethnicity in mortgage approval decisions using recent HMDA data. We estimate logit and probit models and evaluate the sensitivity of estimated marginal effects to the inclusion of race and ethnicity variables.

3. Data Selection

Following Munnell et al. (1996), we retrieve data from the Home Mortgage Disclosure Act (HMDA) database for New York State for the year 2024 (Consumer Financial Protection Bureau, 2024). The resulting data sample contains 112,052 observations. Furthermore, we only consider observations which are classified as either “approved” or “denied” and focus solely on home purchase loans, following Cherian (2014). The reason is the fact that refinancing and home improvement applicants already own a property and therefore the approval decision is less comparable. The variable “interest rate” is excluded as it is only present for applicants who got approved for their loan. Consequently, the models evaluated include explanatory variables grouped into underwriting characteristics, applicant demographics, property attributes, and area-level socioeconomic controls. Together, these factors proxy for the main determinants of the lender’s approval decision (borrower repayment capacity, loan risk, collateral characteristics, and local economic conditions). As such, the model approximates the information available to the lender at the time of assessment. The selected variables are listed and described in Table 1 below:

Table 1: *Variable Description*

Variable	Description
Approved*	Binary variable equal to 1 if the mortgage application was approved and 0 otherwise
Log applicant Income	Natural logarithm of the applicant’s gross annual income (thousands of USD)
Log loan amount	Natural logarithm of the loan amount requested (thousands of USD)

Loan-to-value ratio (LTV)	Ratio of loan amount to property value (%)
Debt-to-income category (DTI)*	Applicant's DTI ratio grouped in discrete categories (<20%, 20-30%, 30-36%, 36-50%, 50-60%, >60%)
Applicant age category*	Age group of primary applicant (<25, 25-34, 35-44, 45-54, 55-64, 65-74, >74)
Applicant sex*	Reported applicant sex (Male, Female, Joint)
Number of housing units*	Total residential units in the collateral property (single-unit, multi-unit (2+))
Dwelling type*	Classification of financed property (site-built, manufactured single-family housing)
Census tract income relative to MSA	Ratio of the median family income in the applicant's census tract to the median family income of his metropolitan statistical area (MSA) (%)
Median income of census tract	Median family income in the MSA where the property is located (thousands of USD)
Race*	White, Black or African American, Asian, American Indian or Alaska Native, Hawaiian or Other Pacific Islander, 2 or more minority races, Other/Not reported
Ethnicity*	Not Hispanic or Latino, Hispanic or Latino, Other/Not reported

*Categorical Variable

Observations with missing or unreasonable values (loan amount and median income of census tract) are trimmed at the 99th percentile, or in case of categorical variables – dropped. The categorical variable “number of housing units” is grouped into binary indicators to improve estimation stability. While previous studies examine a larger set of explanatory variables, either through extensive variable selection within HMDA data or through third-party data (e.g. Cherian, 2014; Munnell et al., 1996; Popick, 2022), this analysis uses a deliberately restricted specification. The purpose is to assess whether differences in mortgage approval outcomes persist even after conditioning solely on a core set of decision-driving variables.

Table 2: Summary Statistics

Variable	Observations	Mean	SD	Min	Max
Approved* (= 1)	98,837	0.88	-	0	1
Applicant Income (in 000s)	112,052	194.65	466.84	1	102,773
Loan Amount (in 000s)	112,052	423.20	316.83	5	2,125
Loan-to-Value ratio (LTV)	112,052	79.48	17.15	0.06	103.41
Median income of census tract	112,052	110.49	23.84	0	156.20
Census tract income to MSA %	112,052	114.92	49.98	0	296.00
Sex Female*	41,284	0.37	-	0	1
American Indian/Alaska*	353	0.00	-	0	1
Asian*	14,891	0.13	-	0	1

Black or African American*	8,168	0.07	-	0	1
Native Hawaiian/ Other*	156	0.00	-	0	1
Hispanic or Latino*	10,801	0.10	-	0	1

*Categorical Variable

Table 2 above presents summary statistics of key selected variables (DTI, age, number of housing units, and dwelling type are omitted for readability). It is evident that the unconditional approval rate for home purchase loans in our sample is 88%. Furthermore, the average loan amount is more than twice the average applicant income, a ratio that has increased over the years (Munnell et al., 1996). On average, applicants borrow around 80% of the property value, 37% of applicants are female, and around 30% of applicants come from a minority group.

4. Methodology

Previous studies on the topic of whether ethnicity and race affect mortgage approval probabilities rely on either logit (e.g. Cherian, 2014; Munnell et al., 1996; Popick, 2022) or probit models (Doviak and MacDonald, 2012). Following the literature, in this paper we estimate both logit and probit specifications and compare the resulting marginal effects. As in Cherian (2014), we specify a latent variable (“random utility”) model, which represents a linear function of standard characteristics considered by the lender in the approval decision, such as credit-related (“underwriting”), individual, property and neighbourhood controls. Since mortgage approval is observed as a binary outcome, it is common to model the decision as resulting from an underlying latent utility structure. In this latent framework, the lender and the borrower are assumed to choose the alternative that maximizes their respective utility (Cherian, 2014). We use the following random utility model formulation (Equation (0)):

$$Approved_i^* = \alpha + \beta' U_i + \gamma' A_i + \delta' P_i + \theta' G_i + \omega' R_i + \varepsilon_i, \quad (0)$$

$$Approved_i = 1\{Approved_i^* > 0\}$$

where ε_i is assumed to follow either a logistic (Λ) or a standard normal (Φ) distribution. Therefore, the probability of mortgage loan approval ($Pr(Approved_i = 1)$) can be modeled the following way:

$$Pr(Approved_i = 1) = F(\alpha + \beta' U_i + \gamma' A_i + \delta' P_i + \theta' G_i) \quad (1)$$

where $F(.)$ denotes either a logistic cumulative distribution function (logit) or a standard normal cumulative distribution function (probit), whose parameters are estimated by maximum likelihood. The vector U_i contains the underwriting controls (log applicant income, log loan amount, loan-to-value ratio, and debt-to-income category) - continuous variables are log-transformed to reduce right skewness and the influence of extreme values. Vector A_i includes applicant characteristics (applicant age category and applicant sex), P_i contains property characteristics (number of housing units and dwelling type), and G_i contains area-level controls (census tract income relative to MSA

and MSA median family income). As such, Equation (1) represents the restricted model, which excludes race and ethnicity controls.

To test whether the probabilities of approval differ systematically with the applicant's race and ethnicity, we explore an extended model specification, Equation (2):

$$\Pr(\text{Approved}_i = 1) = F(\alpha + \beta'U_i + \gamma'A_i + \delta'P_i + \theta'G_i + \omega'R_i) \quad (2)$$

where the vector R_i contains race and ethnicity dummy variables (equal to 0 if applicant i is White/Non-Hispanic or Latino). The comparison between Equations (1) and (2) is therefore the basis for our sensitivity analysis, which tests for whether the inclusion of race and ethnicity affect estimated marginal effects and overall model fit.

To establish inference from the regression results of a binary response model, we report marginal effects, similar to Bostic (1996), Cherian (2014) and Doviak & MacDonald (2012). Average Marginal Effects (AMEs) are computed as the mean of individual partial effects and are interpreted as the average change in the probability of mortgage approval associated with a one-unit increase in the independent variable (if continuous). Formally, for observation i , the marginal effect of a continuous variable x_{ik} is :

$$ME_{ik} = \frac{\partial \Pr(\text{Approved}_i = 1 | x_i)}{\partial x_{ik}} = f(X_i' \hat{\beta}) \hat{\beta}_k$$

where $f(\cdot)$ is the probability density function (the derivative of $F(\cdot)$) of either logistic or normal distribution. X_i denotes the full vector of regressors for observation i , and x_{ik} denotes its k -th element. The AME is then:

$$AME_k = \frac{1}{N} \sum_{i=1}^N f(X_i' \hat{\beta}) \hat{\beta}_k$$

In the case of a categorical variable, the marginal effect is interpreted as the average change in probability of mortgage approval when applicant i belongs to a given category relative to the reference group. For categorical variables:

$$AME_k = \frac{1}{N} \sum_{i=1}^N [F(X_i' \hat{\beta} | x_{ik} = 1) - F(X_i' \hat{\beta} | x_{ik} = 0)]$$

While the models employed in this paper could be misspecified (for example, if unobserved credit scores are correlated with race or ethnicity, as noted by Cherian (2014), and Delis and Papadopoulos (2019), this could lead to biased estimates and false causal interpretation. As such, inference in this study is interpreted only as conditional on the observed controls accounted for. In order to achieve valid statistical inference (conditional on the model being correctly specified), we use heteroscedasticity-robust (White) standard errors, following Popick (2022), which affect only the estimated variances, not the coefficient estimates themselves. The joint significance of race and ethnicity indicators is assessed using Wald test, while their contribution to the model's

performance is evaluated with Likelihood Ratio Index (LRI) and Likelihood Ratio Test (LRT). This is done by comparison of restricted and unrestricted model specifications.

5. Results and Discussion

In binary response models, regression coefficients do not have a direct economic interpretation. Accordingly, Table 3 reports average marginal effects from the unrestricted logit specification. A preliminary inspection indicates that both the unrestricted logit and probit (see Appendix A) models yield average marginal effects of the same direction and similar magnitudes. This suggests that the interpretation of the results is not sensitive to the distributional assumptions we impose on the error term.

Table 3: Average Marginal Effects (Unrestricted Logit)

Variable	AME	Robust SE
Log Income	-0.0216***	(0.0018)
Log Loan Amount	0.0392***	(0.0016)
Loan-to-Value ratio	0.0002**	(0.0001)
Median income of census tract	0.0002***	(0.0000)
Sex Female	0.0017	(0.0018)
Black or African American	-0.0578***	(0.0039)
American Indian/Alaska Native	-0.0642***	(0.0158)
Asian	-0.0504***	(0.0032)
Native Hawaiian/ Other	0.0047	(0.0190)
2 minority races	-0.0960***	(0.0201)
Hispanic or Latino	-0.0496***	(0.0036)

Marginal effects for debt-to-income, age, housing units and dwelling type have been excluded for readability

*p < 0.1, **p < 0.01, ***p < 0.001 (Level of Statistical Significance)

At first glance, the negative statistically significant average marginal effect of log income appears counterintuitive. It is important however to consider the fact that this is the case when controlling for all variables in Equation (2). As such, the sign reversal only occurs once we introduce the debt-to-income categorical variable to the model. Intuitively, increases in income at a fixed debt-to-income ratio imply higher debt levels, which are associated with greater default risk and therefore a lower probability of approval. Therefore, the negative coefficient should be interpreted as a conditional effect, rather than as evidence that higher income reduces the probability of approval. Among underwriting controls, log loan amount and loan-to-value ratio exhibit positive and statistically significant effects on approval probabilities at the 1% and 5% levels respectively. Although the AME of loan-to-value ratio appears small, it represents the effect of a one percentage point increase in leverage and its magnitude is therefore not directly comparable to the other underwriting controls.

Furthermore, sex does not appear to have a statistically significant average marginal effect at any level. This however is not the case with ethnicity and race, with the only exception being Hawaiian or Other Pacific Islander. Nevertheless, it should be noted that it is the least represented minority group in our data sample, which could make inference less reliable. It can also be seen that being classified as any other race or ethnicity considered in this study has a statistically significant negative average marginal effect at the 1% level, and suggests lower approval probabilities relative to the reference group (White). The results coincide with previous research on the topic (Munnell et al., 1996; Cherian, 2014; Popick, 2022; Bostic, 1996),

To further examine the robustness of our results, we compare the Likelihood Ratio Index across specifications. The restricted (see Appendix B) and unrestricted logit models yield values of 0.202 and 0.211, respectively, while the restricted probit model yields 0.201. This suggests that including the race and ethnicity variables in the model improves model fit. Additionally, the Likelihood Ratio tests strongly reject the null hypothesis that race and ethnicity jointly have no explanatory power for mortgage approval outcomes. We observe that the improvement of model fit is significant for both logit (LR=779.78, df=8, $p < 0.001$) and probit (LR=781.52, df=8, $p < 0.001$) specifications. This suggests that, indeed, approval probabilities differ systematically across racial and ethnic groups, even after conditioning on common underwriting, applicant, property and area controls, consistent with prior research. This is further confirmed by the Wald Test, where we observe strong joint statistical significance of race and ethnicity coefficients in both logit ($\chi^2(8)=798.1$, $p < 0.001$) and probit ($\chi^2(8)=786.5$, $p < 0.001$) models.

Given the statistically significant AMEs and robustness checks across logit and probit, we reject our null hypothesis that conditional on observed underwriting, applicant, property, and area characteristics, race and ethnicity have no effect on mortgage approval probabilities. Attention should be drawn to the fact that conclusions from the results of this study should not be understood as causal, but rather conditional on the covariates included in the models. The unconditional approval rates by race and ethnicity are presented in Appendices C and D respectively. As mentioned earlier, correlation between unobserved creditworthiness characteristics (such as credit scores) and included regressors could induce omitted variable bias. This would make our estimates biased and inconsistent. Under correct model specifications and exogeneity conditions, however, the maximum likelihood estimation yields consistent and asymptotically efficient coefficient estimates. Consequently, the robustness of our findings is dependent on the validity of the model specifications used, as well as the adequacy of observable information used to estimate the lender's decision-making.

6. Conclusion

This study provided an empirical application of binary response models to HMDA mortgage approval data for New York State for the year 2024. We estimated the average marginal effects of racial and ethnicity variables on credit approval probabilities and found statistically significant negative conditional effects for most racial and ethnic minority groups. Our results confirm

findings from previous research, that minority groups tend to have lower access to mortgage lending (Cherian, 2014; Bostic, 1996; Doviak and MacDonald, 2012; Delis and Papadopoulos, 2019; Munnell et al., 1996; Popick, 2022).

As mentioned earlier, failure of the exogeneity assumption would render our estimates biased and inconsistent. Consequently, drawing concrete conclusions regarding the presence of discrimination in mortgage lending, would require more extensive data and a more complete set of decision variables considered by lenders at the time of decision. In addition, if it is the case, as Munnell et al. (1996) suggest, that minority applicants are discouraged during the pre-screening process, arriving at accurate results would require a different approach to estimation.

Since this study, as well as much of the existing literature, focuses on regions within the United States, it would be interesting to examine whether European cities exhibit similar dynamics. Furthermore, investigating these effects using panel data and lender fixed effects (although computationally intensive) would provide insights into the development of such disparities through time and across different geographical regions.

Appendix

Appendix A: Average Marginal Effects (Unrestricted Probit)

Variable	AME	Robust SE
Log Income	-0.0230***	(0.0018)
Log Loan Amount	0.0391***	(0.0017)
Loan-to-Value ratio	0.0001*	(0.0001)
Median income of census tract	0.0002***	(0.0000)
Sex Female	0.0022	(0.0018)
Black or African American	-0.0592***	(0.0039)
American Indian/Alaska Native	-0.0663***	(0.0162)
Asian	-0.0482***	(0.0031)
Native Hawaiian/ Other	0.0047	(0.0192)
2 minority races	-0.0962***	(0.0200)
Hispanic or Latino	-0.0495***	(0.0035)

Marginal effects for debt-to-income, age, housing units and dwelling type have been excluded for readability

*p < 0.1, **p < 0.01, ***p < 0.001 (Level of Statistical Significance)

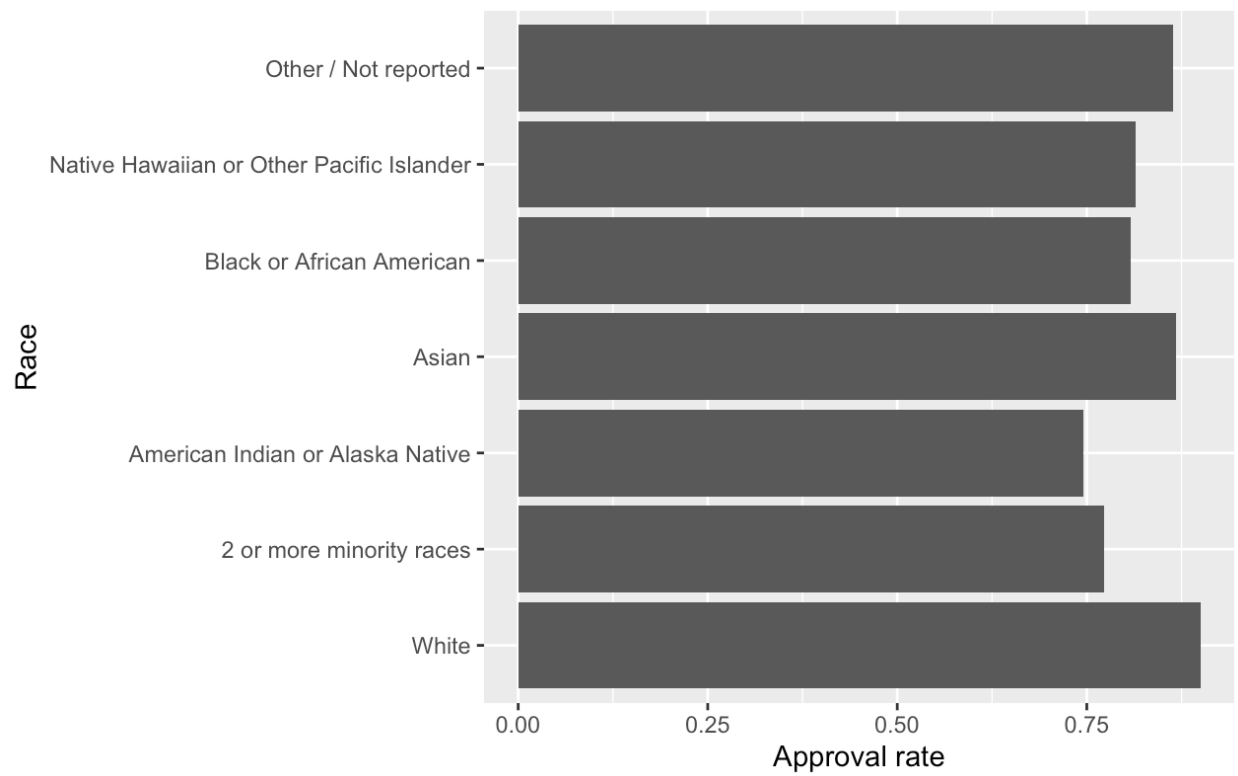
Appendix B: Average Marginal Effect Comparison (Restricted Logit and Probit)

Variable	AME		Robust SE	
	Logit	Probit	Logit	Probit
Log Income	-0.0186***	-0.0196***	(0.0018)	(0.0018)
Log Loan Amount	0.0358***	0.0358***	(0.0016)	(0.0017)
Loan-to-Value ratio	0.0002**	0.0001*	(0.0001)	(0.0001)
Median income of census tract	0.0001*	0.0001**	(0.0000)	(0.0000)
Sex Female	-0.0007	-0.0004	(0.0018)	(0.0018)

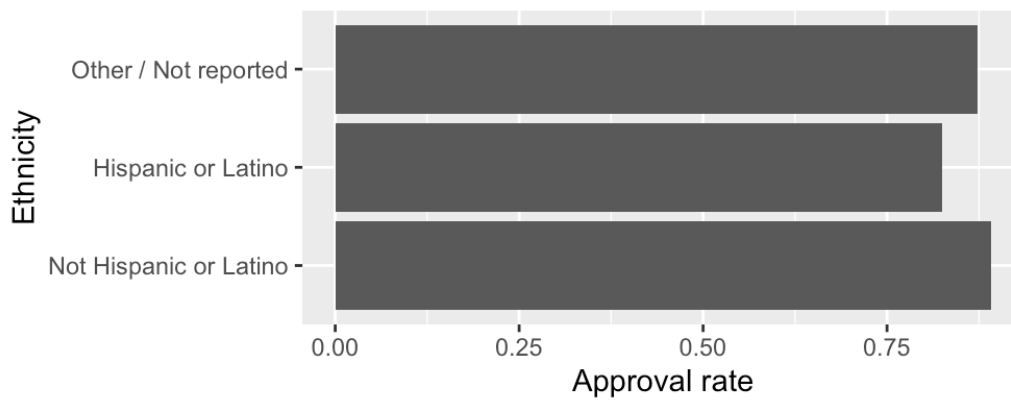
Marginal effects for debt-to-income, age, housing units and dwelling type have been excluded for readability

*p < 0.1, **p < 0.01, ***p < 0.001 (Level of Statistical Significance)

Appendix C: *Unconditional Mortgage Approval Rate by Applicant's Race*



Appendix D: *Unconditional Mortgage Approval Rate by Applicant's Ethnicity*



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